# Med-SkillBench: Evaluating Medical Agent Skills at Scale

Anonymous ACL submission


## Abstract

LLM agents increasingly depend on external skills to perform medical tasks, yet existing benchmarks measure only end-to-end task success—not whether the agent correctly executes the skill specification it was given. We introduce MED-SKILLBENCH, the first large-scale skill-level execution benchmark for medical AI agents, comprising 1,462 skills spanning eight medical domains—from bioinformatics pipelines to clinical reasoning guides—evaluated in isolated sandboxes with a mandatory read-first protocol, eight-dimensional binary scoring, and a novel Specification Groundability Index (SGI) that decomposes execution quality into core fidelity and auxiliary quality components. Our evaluation reveals a systematic *read-but-not-use* (RBU) gap: agents invoke skills at 96.2% but complete specified tasks at only 74.3%, a 22-percentage-point specification grounding failure that decomposes into a dominant semantic grounding component ($\delta_{\text{sem}} = 0.242$) and a negative execution completion component ($\delta_{\text{exec}} = -0.023$), indicating that agents often bypass skill specifications entirely via parametric knowledge. Stratifying by skill interface type shows that guide-style skills (no executable tool) exhibit the largest gap (0.269), while tool-based skills show substantially smaller gaps (0.069–0.111), identifying description groundability—not environment dependency—as the primary bottleneck. Skill-augmented evaluation on 12 medical QA benchmarks further demonstrates that providing matched skills improves accuracy by [TODO: X] percentage points on average, with tool-based skills yielding substantially larger gains than guide-style descriptions. These findings establish description groundability as the primary bottleneck for medical agent skills, quantify the practical value of skill specifications, and provide actionable diagnostic signal for future skill development.


## 1 Introduction

Large language model (LLM) agents increasingly rely on external *skills*—structured tool descriptions that extend model capabilities beyond parametric knowledge—to perform domain-specific tasks. Recent work demonstrates that curated skill libraries substantially improve agent performance: **?** report an average gain of 16.2 percentage points across domains, with the medical domain exhibiting the largest improvement at 51.9 percentage points. This finding underscores the critical role of skill quality in high-stakes settings such as clinical decision support, biomedical literature analysis, and drug safety assessment.

Despite this promise, medical skills present unique evaluation challenges that existing benchmarks do not address. Medical skill specifications encode domain-specific safety boundaries, clinical workflow constraints, and a fundamental *reasoning–execution duality*: some skills wrap executable tools (BLAST, RDKit, clinical calculators), while others provide purely reasoning-based guidance (differential diagnosis protocols, evidence synthesis frameworks). Both modes are clinically legitimate but require fundamentally different quality criteria.

**Task-level evaluation obscures skill-level failures.** Existing medical agent benchmarks evaluate end-to-end task success (**???**) but do not isolate whether the agent correctly *executed* the skill specification it was given. An agent may answer a clinical question correctly by ignoring a poorly written drug interaction checker, or it may fail a genomic analysis task despite faithfully following a specification that omits critical parameter constraints. Task-level metrics conflate these fundamentally different failure modes, leaving skill developers without actionable feedback on *where* in the specification–execution chain failures originate.

MED-SKILLBENCH addresses this gap through

**[FIGURE PLACEHOLDER]**
*Figure 1 (Teaser): the* RBU *gap. Left panel—agents read* `SKILL.md` *at 96.2% invocation. Right panel—only 74.3% complete the specified task. The 22-pp wedge between the two represents specification grounding failure.*

Figure 1: The RBU gap: agents read skill specifications (96.2%) yet only complete skill-grounded tasks at 74.3%, exposing a 22-percentage-point specification grounding failure.

systematic skill-level evaluation that treats skill descriptions as first-class agent interfaces whose execution quality can be independently measured, decomposed across eight diagnostic dimensions, and stratified by medical domain.[1]

**Contribution 1: MED-SKILLBENCH.** We introduce the first large-scale skill-level execution benchmark for medical AI agents. MED-SKILLBENCH aggregates 1,462 skills from three open-source medical skill repositories and evaluates each in an isolated sandbox environment. A mandatory *read-first* protocol ensures the agent must parse the `SKILL.md` specification before attempting execution. An LLM-as-judge scores each execution trajectory along eight binary dimensions—*skill invocation*, *correct usage*, *task completion*, *tool execution success*, *output quality*, *reasoning quality*, *efficiency*, and *trajectory quality*—providing fine-grained diagnostic signal rather than a single pass/fail label.

**Contribution 2: RBU gap quantification.** Using MED-SKILLBENCH, we identify and quantify a systematic *read-but-not-use* (RBU) gap: agents invoke skills at a rate of 96.2% but complete the specified tasks at only 74.3%, yielding a 22-percentage-point gap. This gap represents a *specification grounding failure*—the model recognizes what a skill describes but cannot reliably translate that description into correct execution. Stratifying by `tool_type` reveals that the gap is strongly modulated by skill interface design: guide-style skills (no executable tool) exhibit an RBU gap of 0.269, while tool-based skills show substantially smaller gaps (CLI: 0.106; Python: 0.099; mixed: 0.069; R: 0.111). This finding shifts the diagnosis from "models cannot use tools" to "models struggle most when skill descriptions lack executable grounding." A formal two-stage decomposition reveals that the gap is overwhelmingly driven by semantic grounding failure ($\delta_{\text{sem}} = 0.242$) rather than execution failure ($\delta_{\text{exec}} = -0.023$), with the negative execution component indicating that models sometimes succeed by bypassing the skill specification entirely.

**Contribution 3: Skill-augmented QA evaluation.** We evaluate whether skill availability measurably improves model performance on real medical benchmark questions. By pairing validated skill–question matches with a controlled with/without-skill ablation across multiple executor models, we quantify the accuracy uplift attributable to skill specifications and show that the uplift is strongly modulated by skill interface type—consistent with the RBU gap analysis. [TODO: Report uplift magnitude and statistical significance.]

Our work is organized around three research questions:

**RQ1:** Do medical agent skills exhibit a systematic read-but-not-use gap between specification comprehension and execution?

**RQ2:** Is the gap primarily attributable to environment dependency or to description-level grounding failure?

**RQ3:** Does skill availability improve model accuracy on real medical QA benchmarks, and is the uplift modulated by skill execution quality?

**Positioning.** Following the taxonomy of **?**, MED-SKILLBENCH operates primarily at the *Skill Deployment* tier (T2), evaluating skills post-authoring. It secondarily informs *Skill Development* (T1) by identifying which description patterns yield higher groundability. The benchmark requires no model training and is applicable to closed-weight models and resource-constrained settings alike.

The remainder of this paper is structured as follows. §2 reviews related work. §3 details the MED-SKILLBENCH evaluation protocol and the skill-augmented QA evaluation design. §4 reports experimental results, including skill-augmented performance analysis, and discusses implications. §5

[1] Code, data, and evaluation artifacts are available at https://anonymous.4open.science/r/Med-SkillBench-TODO.

concludes.

## 2 Related Work

### 2.1 Agent Skills and Skill Benchmarks

Skills extend language-model agents with reusable procedural knowledge: task descriptions, invocation conditions, parameter schemas, and multi-step workflows. **?** introduce SKILLSBENCH, evaluating 86 tasks across 11 domains. Curated skills boost agent performance by an average of 16.2 percentage points, with the medical domain showing the largest gain (+51.9 pp), while self-generated skills yield near-zero improvement. **?** scale the effort to over 200K skills with SKILLNET, proposing multi-dimensional *static* evaluation along Safety, Completeness, Executability, Maintainability, and Cost-awareness.

Both contributions establish that skill quality matters. However, neither grounds its evaluation in *execution traces*: SKILLSBENCH measures task-level delta, not whether the skill was actually invoked and followed; SKILLNET scores interface quality without running the skill in an isolated sandbox. MED-SKILLBENCH bridges this gap by evaluating 1,462 medical skills (917 Tier-A) at the execution level, quantifying the RBU gap between reading a `SKILL.md` and correctly executing it.

### 2.2 Medical and Biomedical Agent Benchmarks

Recent medical benchmarks have diversified the evaluation surface for language-model agents. **?** present MEDAGENTBENCH, comprising 300 FHIR-compliant EHR tasks that test interactive clinical actions. **?** release HEALTHBENCH with 5,000 multi-turn health dialogues scored by 262 physician-authored rubrics. In biomedical research, **?** propose LAB-BENCH, covering 2,400+ biology research questions spanning LitQA, DBQA, and ProtocolQA. **?** introduce the HUMANITY'S LAST EXAM (HLE), a 2,500-question expert-level benchmark published in *Nature*, whose Biomedicine subset—together with LAB-BENCH—has proven effective for evaluating biomedical agents (**?**). Interactive tool-use benchmarks such as $\tau$-bench (**?**) and TOOLSANDBOX (**?**) further formalize stateful, multi-turn evaluation with programmatic verification. **?** contribute MEDAGENTGYM, framing medical agent training as a reinforcement learning problem over clinical environments.

Table 1: Comparison of MED-SKILLBENCH with related work. *Eval.*: task-level (T) or skill-level (S); parenthetical indicates execution-based (E) or static (S) evaluation. *Opt. Target*: what is optimized. *Scale*: number of evaluation items.

| Work | Eval. | Domain | Opt. Target | Scale |
|---|---|---|---|---|
| SkillsBench | T (S) | General | — | 86 |
| SkillNet | S | General | — | 200K+ |
| MedAgentBench | T (E) | Clinical | — | 300 |
| HealthBench | T (E) | Health | — | 5,000 |
| LAB-Bench | T (S) | Biomed. | — | 2,400+ |
| HLE | T (S) | Multi | — | 2,500 |
| EasyTool | — | General | Tool desc. | — |
| Play2Prompt | — | General | Tool desc. | — |
| Trace-Free+ | — | General | Tool desc. | — |
| **MED-SKILLBENCH** | **S (E)** | **Medical** | **—** | **1,462** |

These benchmarks assess *task-level* success. MED-SKILLBENCH operates at a complementary *skill level*: given a specific `SKILL.md`, does the agent read, invoke, and correctly follow the skill specification?

### 2.3 Tool and Skill Description Optimization

A growing body of work shows that how tools are *described* is itself a performance lever. **?** compress verbose, heterogeneous tool documentation into concise, standardized instructions (EASYTOOL). **?** let agents interactively explore tools to auto-generate high-quality usage demonstrations (PLAY2PROMPT). **?** address settings where execution traces are unavailable, using curriculum transfer from trace-rich to trace-free environments to generalize tool-description rewriting (TRACE-FREE+). The ReAct paradigm (**?**) further underscores the tight coupling between reasoning traces and tool invocation fidelity.

These methods target *generic* tool descriptions—typically API signatures or short docstrings. Medical skills are substantially more complex: they encode domain terminology, parameter schemas, safety boundaries, dependency constraints, and multi-step clinical workflows within a single `SKILL.md`. Our benchmark provides the diagnostic signal needed to evaluate such optimization methods on full-length, structured medical skill documents.

**Positioning.** Table 1 summarizes how MED-SKILLBENCH relates to prior work across five key dimensions.

## 3 Method

We formalize the skill-grounded execution problem before describing the benchmark design and the skill-augmented evaluation protocol.

### 3.1 Problem Formulation

**Definition 3.1** (Skill Specification)**.** A *skill specification* is a tuple $s = (\texttt{desc}, \texttt{params}, \texttt{iface}, \texttt{dom})$ where: $\texttt{desc} \in \Sigma^*$ is a natural-language description document (the `SKILL.md` content); $\texttt{params} = \{(p_j, T_j, C_j)\}_{j=1}^{m}$ is a parameter schema with type $T_j$ and constraint set $C_j$; $\texttt{iface} \in \mathcal{I} = \{\texttt{cli}, \texttt{python}, \texttt{R}, \texttt{mixed}, \texttt{none}\}$ is the tool interface type; and $\texttt{dom} \in \mathcal{D}$ is the medical domain label from a fixed taxonomy $\mathcal{D}$. We write $\mathcal{S}$ for the full skill catalog and $\mathcal{S}_A \subseteq \mathcal{S}$ for the Tier-A subset surviving quality filtering.

**Definition 3.2** (Agent Execution)**.** An *agent* $\mathcal{M}$ is a stochastic function that, given a task description $x \in \mathcal{X}$ and a skill specification $s \in \mathcal{S}$, samples a bounded execution trajectory:

$$\tau \sim \mathcal{M}(x, s), \quad \tau = (a_1, o_1, \ldots, a_T, o_T), \quad T \leq T_{\max} \tag{1}$$

where $a_t \in \mathcal{A}$ is an agent action (tool call, code execution, reasoning step, file read) and $o_t \in \mathcal{O}$ is the environment observation returned after $a_t$. The trajectory is bounded by $T_{\max} = 40$ iterations and a wall-clock timeout $\Delta t_{\max} = 300$s.

**Remark (Deterministic Approximation).** In practice, all experiments use temperature 0 (greedy decoding), making $\mathcal{M}$ effectively deterministic for a given $(x, s)$. We retain the stochastic formulation for generality, but all reported metrics are point estimates from single runs.

**Definition 3.3** (Evaluation Function)**.** The *evaluation function* $E$ maps a (trajectory, skill, task) triple to an eight-dimensional binary quality vector:

$$E : (\mathcal{A} \times \mathcal{O})^{\leq T_{\max}} \times \mathcal{S} \times \mathcal{X} \longrightarrow \{0,1\}^8, \quad E(\tau, s, x) = \boldsymbol{y}. \tag{2}$$

The eight dimensions are partitioned into **core fidelity** $\boldsymbol{y}^{\text{core}} = (y_1, y_2, y_3)$ and **auxiliary quality** $\boldsymbol{y}^{\text{aux}} = (y_4, \ldots, y_8)$:

| Index | Dimension | Interpretation |
|---|---|---|
| $y_1$ | `skill_invoked` | Agent reads/references the specification |
| $y_2$ | `skill_used_correctly` | Actions conform to specification semantics |
| $y_3$ | `task_completion` | Final output satisfies the task objective |
| $y_4$ | `tool_execution_success` | $\geq 1$ tool call returns verified output |
| $y_5$ | `output_quality` | Domain-appropriate quality of final output |
| $y_6$ | `reasoning_quality` | Coherent, non-circular reasoning chain |
| $y_7$ | `efficiency` | Reasonable iteration count, no retry loops |
| $y_8$ | `trajectory_quality` | Purposeful progression and error recovery |

**Remark (Soft Dependency Structure).** The core dimensions encode a *soft* dependency tendency: correct usage ($y_2{=}1$) typically presupposes invocation ($y_1{=}1$), and task completion ($y_3{=}1$) typically presupposes correct usage ($y_2{=}1$). However, this is not a strict logical entailment. Empirically, $\bar{y}_3 > \bar{y}_2$ is possible (and observed: $\bar{y}_3 = 0.743 > \bar{y}_2 = 0.720$ in Table 2), because an agent can complete a task via parametric knowledge even when it fails to follow the skill specification correctly. This is precisely the phenomenon MED-SKILLBENCH is designed to detect: successful task completion that bypasses the skill specification should not be counted as successful skill grounding.

The benchmark design is detailed in Section 3.1.

### 3.2 MED-SKILLBENCH: Skill-Level Execution Benchmark

Existing medical agent benchmarks evaluate end-to-end task completion (**??**) or library-level tool retrieval (**??**), but neither setup isolates the contribution of an individual skill's *description quality* to execution success. MED-SKILLBENCH fills this gap with four design requirements: (1) skill-level isolation—each skill is evaluated independently; (2) execution fidelity—scores reflect real tool invocation in a sandboxed agent loop; (3) failure attribution—the scoring scheme distinguishes where in the execution chain a failure occurs; (4) reusability—the same protocol can be applied to different description variants or across models.

**Skill Collection and Quality Filtering.** We harvest skills from three open-source medical skill repositories covering genomics, proteomics, single-cell analysis, chemoinformatics, bioinformatics, clinical decision support, and medical research tooling. De-duplication yields **1,462** unique skills, each defined by a `SKILL.md` specifying name, description, parameter schema, and optionally usage examples and safety notes. Before agent execution, a five-dimensional binary static pre-scan checks: (i) `can_get_credentials`, (ii) `content_complete`, (iii) `dependencies_resolvable`, (iv) `has_clear_io`, (v) `has_documentation`. The surviving **917 Tier-A skills** form the evaluation set.

**Isolated Evaluation Protocol.** Each skill is evaluated inside an independent temporary workspace containing only the skill's own artifacts. No other skills, shared registries, or caches are present.

The agent operates within a bounded AgentLoop of at most 40 iterations with a 300-second wall-clock timeout. To ensure the agent reads the description before acting, we prepend a *mandatory read-first* instruction to every prompt, making the `skill_invoked` dimension an interpretable signal: $y_1 = 0$ means the agent ignored the description even when directed to read it.

**Task Construction.** For each skill we generate a diagnostic synthetic task matching the skill's declared purpose. Tasks require actual tool execution rather than planning, with concrete completion criteria. A separate LLM generates each task, enforcing **role separation**: task generator, executor, and judge are distinct model calls.

**Eight-Dimensional Binary Scoring.** Each evaluation produces a vector $\boldsymbol{y} = (y_1, \ldots, y_8) \in \{0,1\}^8$ as defined in **??**. We define three outcome tiers:

$$\textsc{Pass}(s) = \mathbf{1}\big[y_1{=}1 \ \wedge\ y_2{=}1 \ \wedge\ y_3{=}1\big], \quad (3)$$

$$\textsc{Gold}(s) = \mathbf{1}\left[\textstyle\sum_{i=1}^{8} y_i = 8\right], \quad (4)$$

$$\textsc{Silver}(s) = \textsc{Pass}(s) \cdot \big(1 - \textsc{Gold}(s)\big). \quad (5)$$

Gold requires full execution quality across all dimensions. Silver indicates core fidelity without full auxiliary quality—the skill was correctly invoked and the task completed, but execution artifacts (tool failures, inefficiency, trajectory issues) remain. Denoting the sets of skills achieving each tier as $\mathbb{G}$, $\mathbb{S}$, and $\mathbb{F}$ (core-failure), we have $\mathbb{G} \subset \mathbb{P}$, $\mathbb{S} \subset \mathbb{P}$, $\mathbb{G} \cap \mathbb{S} = \varnothing$, and the *disjoint* partition $\mathbb{P} = \mathbb{G} \sqcup \mathbb{S} \sqcup \mathbb{F}$. Binary dimensions yield unambiguous pass/fail decisions, avoiding calibration difficulties of Likert-scale scoring across heterogeneous medical domains.

**The Read-But-Not-Use Gap.** We formalize the gap between invocation and completion. The *aggregate* RBU *gap* is:

$$\Delta_{\mathrm{RBU}}(\mathcal{S}, \mathcal{M}) = \bar{y}_1 - \bar{y}_3, \quad \bar{y}_k = \tfrac{1}{|\mathcal{S}|}\textstyle\sum_{s\in\mathcal{S}} y_k^{(s)}. \quad (6)$$

**Proposition 3.4** (Two-Stage Decomposition of the RBU Gap)**.** *The* RBU *gap decomposes additively into two mechanistically distinct failure stages:*

$$\Delta_{\mathrm{RBU}} = \underbrace{(\bar{y}_1 - \bar{y}_2)}_{\delta_{\text{sem}}} + \underbrace{(\bar{y}_2 - \bar{y}_3)}_{\delta_{\text{exec}}}, \quad (7)$$

*where $\delta_{\text{sem}} = \bar{y}_1 - \bar{y}_2$ is the* ***semantic grounding gap*** *(specification comprehension→action fidelity) and $\delta_{\text{exec}} = \bar{y}_2 - \bar{y}_3$ is the* ***execution completion gap*** *(action→outcome fidelity).*

*Proof.* Telescoping: $\bar{y}_1 - \bar{y}_3 = (\bar{y}_1 - \bar{y}_2) + (\bar{y}_2 - \bar{y}_3)$. □

**Remark (Sign Interpretation).** Both $\delta_{\text{sem}}$ and $\delta_{\text{exec}}$ can take negative values. In the overall corpus, $\delta_{\text{exec}} = 0.720 - 0.743 = -0.023 < 0$, meaning task completion slightly *exceeds* correct skill usage. This arises when agents complete tasks by falling back to parametric knowledge rather than following the skill—they "pass the exam without reading the textbook." Far from being a pathology, negative $\delta_{\text{exec}}$ is a key diagnostic signal: it indicates the task is solvable without the skill, which is important context for interpreting downstream uplift. The semantic grounding gap $\delta_{\text{sem}} = 0.962 - 0.720 = 0.242$ accounts for more than 100% of the aggregate RBU gap, confirming that the dominant bottleneck is specification-to-action translation.

**Remark (Diagnostic Utility).** Consider two skill categories with identical $\Delta_{\mathrm{RBU}} = 0.22$: **Category A** (guide-style) has $\delta_{\text{sem}} = 0.18$, $\delta_{\text{exec}} = 0.04$—the bottleneck is specification comprehension. **Category B** (tool-based) has $\delta_{\text{sem}} = 0.04$, $\delta_{\text{exec}} = 0.18$—the bottleneck is execution. The aggregate $\Delta_{\mathrm{RBU}}$ cannot distinguish these; the decomposition can, and it directly prescribes different interventions (improve descriptions vs. improve tool environments).

For interface type $\iota \in \mathcal{I}$, let $\mathcal{S}_\iota = \{s \in \mathcal{S} : s.\mathtt{iface} = \iota\}$. The *interface-conditioned* RBU gap and its decomposition are:

$$\Delta_{\mathrm{RBU}}^{(\iota)} = \bar{y}_1^{(\iota)} - \bar{y}_3^{(\iota)} = \delta_{\text{sem}}^{(\iota)} + \delta_{\text{exec}}^{(\iota)}, \quad (8)$$

where all means are taken over $\mathcal{S}_\iota$.

**Hypothesis 1 (Interface Grounding).** *Structured tool interfaces reduce the* RBU *gap primarily by shrinking the semantic grounding component:*

$$\delta_{\text{sem}}^{(\text{none})} \gg \delta_{\text{sem}}^{(\iota)}, \quad \forall\, \iota \in \{\mathtt{cli}, \mathtt{python}, \mathtt{R}, \mathtt{mixed}\}. \quad (9)$$

When a skill specifies a concrete invocation template (e.g., `blastp -query input.fasta -db nr`), the mapping from description to action is near-deterministic. Guide-style skills require the agent to synthesize multi-step action plans from unconstrained prose, introducing a combinatorially larger space of possible (mis)interpretations.

**Specification Groundability Index.** Beyond the aggregate RBU gap, we introduce a per-skill *Specification Groundability Index* (SGI) that captures the full execution quality profile:

$$\mathrm{SGI}(s, \mathcal{M}) = \underbrace{w_c \cdot \tfrac{1}{3} \textstyle\sum_{i=1}^{3} y_i}_{\text{core fidelity } F_c(s)} + \underbrace{(1{-}w_c) \cdot \tfrac{1}{5} \textstyle\sum_{i=4}^{8} y_i}_{\text{auxiliary quality } Q_a(s)}, \tag{10}$$

where $w_c \in (0, 1)$ controls the emphasis on core fidelity relative to auxiliary quality. We set $w_c$ based on a **dominance principle**: a skill that fails on any core dimension is clinically unusable regardless of auxiliary quality, so core fidelity must dominate. Formally, we require that the *worst-case* core-failing skill always scores below the *worst-case* core-passing skill:

$$\max_{\boldsymbol{y}:\, F_c<1} \mathrm{SGI}(\boldsymbol{y}) \;<\; \min_{\boldsymbol{y}:\, F_c=1} \mathrm{SGI}(\boldsymbol{y}). \tag{11}$$

The left maximum is achieved by a skill with $F_c = 2/3$ and $Q_a = 1$ (one core failure, all auxiliary pass); the right minimum by $F_c = 1$ and $Q_a = 0$ (all core pass, all auxiliary fail):

$$\underbrace{w_c \cdot \tfrac{2}{3} + (1{-}w_c) \cdot 1}_{=1-w_c/3} \;<\; \underbrace{w_c \cdot 1 + (1{-}w_c) \cdot 0}_{=w_c} \quad\Longrightarrow\quad w_c > \tfrac{3}{4} = 0.75. \tag{12}$$

We set $w_c = 0.75$, the tightest value satisfying strict dominance—the *least opinionated* choice guaranteeing that any core-failing skill always ranks below any core-passing skill. We verify robustness to $w_c \in \{0.6, 0.7, 0.75, 0.8\}$ in sensitivity analysis (Appendix G).

**Proposition 3.5** (SGI Decomposition by Domain)**.** *For domain $d \in \mathcal{D}$, the mean* SGI *decomposes as:*

$$\overline{\mathrm{SGI}}_d = w_c \cdot \bar{F}_{c,d} + (1 - w_c) \cdot \bar{Q}_{a,d} \tag{13}$$

This decomposition exposes cross-domain quality patterns: computational domains (bioinformatics) are predicted to show $\bar{F}_{c,d} > \bar{Q}_{a,d}$ (high core fidelity, tool execution failures drag down auxiliary), while clinical reasoning domains show $\bar{Q}_{a,d} > \bar{F}_{c,d}$ (high reasoning quality, but weaker specification adherence). This enables *domain-aware* skill quality diagnosis.

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**Algorithm 1** MED-SKILLBENCH Evaluation Pipeline

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**Require:** skill catalog $\mathcal{S}$, task generator $\mathcal{G}$, executor model $M$, judge model $J$
**Ensure:** per-skill score vectors $\{(\boldsymbol{y}_s, \mathrm{status}_s)\}_{s\in\mathcal{S}}$

1: **for** each skill $s \in \mathcal{S}$ **do**
2: $\quad x \leftarrow \mathcal{G}(s)$ ▷ Generate diagnostic task from `SKILL.md`
3: $\quad$Create isolated workspace with only $s$'s artifacts
4: $\quad$Inject mandatory read-first prompt into agent system message
5: $\quad \tau \leftarrow \textsc{AgentLoop}(M,\, s,\, x)$ ▷ ≤40 iter, 300 s timeout
6: $\quad \boldsymbol{y}_s \leftarrow J(\tau,\, s,\, x) \in \{0,1\}^8$ ▷ 8-dim binary scoring
7: $\quad \mathrm{status}_s \leftarrow \begin{cases} \textsc{Gold} & \text{if } \sum y_i = 8 \\ \textsc{Pass} & \text{if } y_1 \wedge y_2 \wedge y_3 \\ \textsc{Fail} & \text{otherwise} \end{cases}$
8: **end for**
9: **return** $\{(\boldsymbol{y}_s, \mathrm{status}_s)\}_{s\in\mathcal{S}}$

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**Medical Domain Taxonomy.** Medical skills differ fundamentally from generic tool descriptions in three respects. First, they encode *domain-specific safety boundaries*—a drug interaction checker must not silently drop contraindications; a variant annotation pipeline must preserve clinical significance flags. Second, medical skills span a *reasoning–execution duality*: 64.2% of Tier-A skills are guide-style (clinical reasoning, evidence synthesis, study design) with no executable tool interface, while 35.8% invoke computational tools (BLAST, RDKit, samtools). Both modes are clinically legitimate, but they require fundamentally different evaluation: tool-based skills are assessed on execution fidelity, while reasoning-based skills are assessed on semantic correctness. Third, medical skills exhibit a *heterogeneous domain distribution*: the 917 Tier-A skills span eight medical domains—bioinformatics (48.5%), clinical medicine (8.8%), biomedical research (6.0%), drug discovery (3.8%), public health (3.1%), structural biology (2.8%), medical imaging (2.0%), and cross-cutting research support (25.0%). This imbalance reflects the real-world landscape of open-source medical agent tools and motivates domain-stratified evaluation.

### 3.3 Skill-Augmented QA Evaluation Protocol

The benchmark evaluation (Section 3.1) assesses whether agents can execute skill specifications in isolation. A complementary question is whether skill availability actually *improves* model performance on real medical questions. We address this through a controlled with/without-skill ablation.

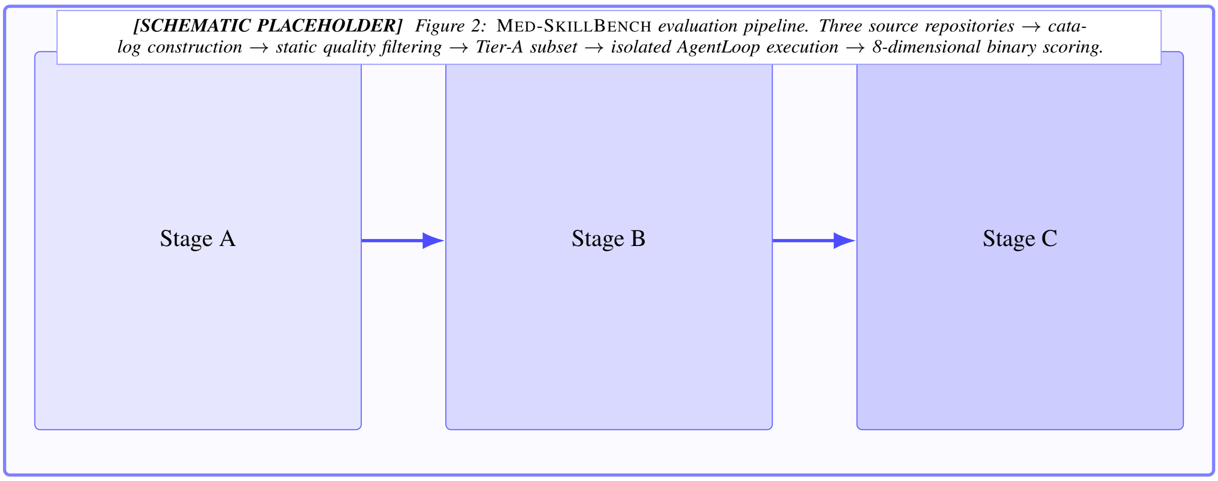


Figure 2: Overview of the MED-SKILLBENCH evaluation pipeline, spanning catalog construction, static quality filtering, isolated agent execution, and 8-dimensional binary scoring.

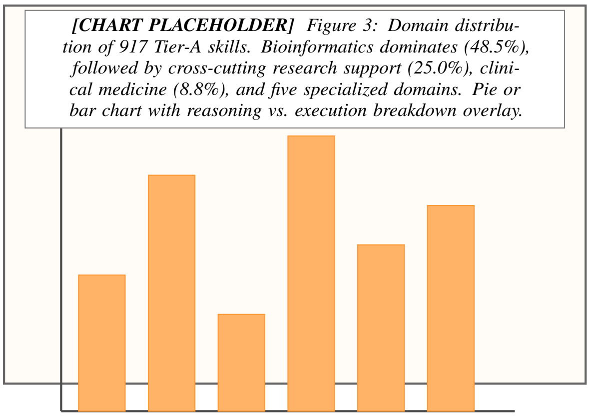


Figure 3: Medical domain distribution of the MED-SKILLBENCH skill catalog. The heterogeneous distribution reflects the real-world landscape of open-source medical agent tools.

**Task Selection.** We link the 917 Tier-A skills to 14,019 questions from 12 established medical QA benchmarks via a three-round matching pipeline: (1) batched LLM matching identifies candidate skill–question pairs; (2) multi-dimensional validation filters for relevance, execution feasibility, and domain alignment; (3) final scoring merges static quality and fitness signals. Let $\mu : \mathcal{Q} \to 2^{\mathcal{S}}$ be the resulting skill-matching function, assigning each question to a (possibly empty) set of relevant skills. We expand to *skill–question pairs*:

$$\mathcal{P}_\mu = \{(x_i, s, g_i) : s \in \mu(x_i),\ \mu(x_i) \neq \varnothing\}.$$

The surviving validated pairs form the ablation evaluation set. [TODO: Report exact number of validated pairs after R2/R3 completion.]

**Dual-Mode Execution.** Each validated pair $(x_i, s, g_i) \in \mathcal{P}_\mu$ is evaluated under two conditions using the same executor model and identical hyperparameters:

- **With-Skill**: The agent's workspace contains the matched `SKILL.md` file alongside the question. The mandatory read-first prompt directs the agent to consult the skill specification.
- **Without-Skill**: The agent receives only the question in an empty workspace, relying entirely on parametric knowledge.

Both conditions use the same bounded AgentLoop (40 iterations, 300-second timeout) to ensure comparability. Define accuracy under the with-skill condition over the pair set:

$$\mathrm{Acc}_{\text{with}}(\mathcal{P}_\mu, \mathcal{M}) = \frac{1}{|\mathcal{P}_\mu|} \sum_{(x_i, s, g_i) \in \mathcal{P}_\mu} \mathbf{1}[\mathcal{M}(x_i, s) = g_i], \tag{14}$$

and accuracy under the without-skill baseline:

$$\mathrm{Acc}_{\text{w/o}}(\mathcal{P}_\mu, \mathcal{M}) = \frac{1}{|\mathcal{P}_\mu|} \sum_{(x_i, s, g_i) \in \mathcal{P}_\mu} \mathbf{1}[\mathcal{M}(x_i, \varnothing) = g_i]. \tag{15}$$

When $|\mu(x_i)| > 1$, the same question $x_i$ appears in multiple pairs; the without-skill prediction $\mathcal{M}(x_i, \varnothing)$ is computed once and reused across all pairs sharing the same $x_i$.

**Answer Evaluation.** For multiple-choice questions, we apply exact option matching. For open-ended and numeric questions, an LLM-as-judge determines correctness against the gold answer, maintaining role separation from the executor model. Each trial produces a binary *correct* label.

**Uplift Metric.** We define the skill uplift as the paired accuracy difference:

$$\Delta_{\text{uplift}}(\mathcal{P}_\mu, \mathcal{M}) = \text{Acc}_{\text{with}} - \text{Acc}_{\text{w/o}}. \quad (16)$$

A positive $\Delta_{\text{uplift}}$ indicates that skill availability improves performance. For each pair $(x_i, s, g_i) \in \mathcal{P}_\mu$, define the *per-pair uplift indicator*:

$$u_{i,s} = \mathbf{1}[\mathcal{M}(x_i, s) = g_i] - \mathbf{1}[\mathcal{M}(x_i, \varnothing) = g_i] \in \{-1, 0, +1\}. \quad (17)$$

Then $\Delta_{\text{uplift}} = \bar{u}$. The *net benefit ratio* $\rho = n_{+1}/(n_{+1} + n_{-1})$ indicates whether skills help more pairs than they hurt ($\rho > 0.5$).

**Connecting Diagnostics to Utility.** The preceding sections establish two independent measurement instruments: the RBU gap with its SGI decomposition (Step 0, Section 3.1), and the with/without-skill ablation (Step 2, this section). We now connect them.

Partition matched pairs by interface type: $\mathcal{P}_\mu^{(\iota)} = \{(x_i, s, g_i) \in \mathcal{P}_\mu : s.\texttt{iface} = \iota\}$. The *interface-conditioned uplift* is:

$$\Delta_{\text{uplift}}^{(\iota)} = \text{Acc}_{\text{with}}^{(\iota)} - \text{Acc}_{\text{w/o}}^{(\iota)}, \quad \iota \in \mathcal{I}. \quad (18)$$

**Proposition 3.6** (Groundability-Modulated Uplift)**.** *If the* RBU *gap reflects genuine specification grounding failure, then interface types with* smaller RBU *gaps should exhibit* larger *uplift:*

$$\Delta_{\text{RBU}}^{(\iota_1)} < \Delta_{\text{RBU}}^{(\iota_2)} \implies \Delta_{\text{uplift}}^{(\iota_1)} \geq \Delta_{\text{uplift}}^{(\iota_2)}. \quad (19)$$

We test this prediction in two ways. First, the aggregate *groundability–uplift correlation*:

$$r_{GU} = \text{Corr}\Big(\{-\Delta_{\text{RBU}}^{(\iota)}\}_{\iota \in \mathcal{I}},\ \{\Delta_{\text{uplift}}^{(\iota)}\}_{\iota \in \mathcal{I}}\Big).$$

A positive $r_{GU}$ validates the "closing the loop" claim. However, with $|\mathcal{I}| = 5$ interface types, this correlation has negligible statistical power ($n = 5$ requires $|r| > 0.878$ for $p < 0.05$). We therefore adopt the following **primary test**: for each skill $s$ with at least $k \geq 5$ matched QA pairs, regress per-skill uplift on SGI:

$$\Delta_{\text{uplift}}^{(s)} = \beta_0 + \beta_1 \cdot \text{SGI}(s) + \epsilon_s. \quad (20)$$

A significantly positive $\hat{\beta}_1$ provides a higher-powered test of the same hypothesis, using individual skills as the unit of analysis. We report both $r_{GU}$ and $\hat{\beta}_1$; the latter is the primary test.

**Statistical Testing.** Because each question is evaluated under both conditions by the same model, we use McNemar's paired test for binary outcomes, supplemented by bootstrap 95% confidence intervals (10,000 resamples, clustered by question index to handle within-question dependence when $|\mu(x_i)| > 1$). Full formalization of the McNemar statistic and bootstrap procedure is provided in **??**. We stratify results by benchmark, `tool_type`, and medical domain to identify where skill augmentation provides the largest benefit.

All evaluation transcripts, scores, and skill–benchmark matching tables are released as a frozen v2 snapshot.[2]

# 4 Experiments

We evaluate MED-SKILLBENCH along three complementary axes: (1) large-scale skill quality assessment via the 8-dimensional evaluation framework (§4.1–§4.3), (2) downstream grounding of skills to real medical benchmark questions (§4.5), and (3) skill-augmented QA evaluation measuring the performance uplift from skill availability (§4.6–§4.7), followed by a discussion of the findings (§4.8).

## 4.1 Experimental Setup

**Models.** We use GPT-5.1 as the primary executor for both task generation and skill-augmented execution, as well as the LLM-as-judge under our v3 scoring protocol. For the skill-augmented QA evaluation (§4.6), we additionally evaluate Claude Sonnet 4.5 to assess cross-model consistency of the skill uplift effect.

**Execution environment.** Each skill evaluation trial runs in an isolated workspace with a 300-second wall-clock timeout and a maximum of 40 agent iterations. The agent operates within a sandboxed nanobot AgentLoop that provides file-system access, shell execution, and Python/R interpreters, but prohibits network access during evaluation. This design ensures that observed tool execution outcomes reflect the skill specification rather than external data retrieval.

**Evaluation protocol.** We employ the 8-dimensional binary scoring framework introduced in Section 3.1. Each dimension produces a

[2] Anonymous access: https://anonymous.4open.science/r/Med-SkillBench-TODO; camera-ready will include a permanent DOI.

binary pass/fail judgment; a skill *passes* overall if all three *core* dimensions (`skill_invoked`, `skill_used_correctly`, `task_completion`) receive a score of 1. A skill achieves *gold* status when all eight dimensions score 1.

**Evaluation conditions.** All results report the **Original-Skill** condition: the agent receives the unmodified `SKILL.md` as authored by the skill developer. We additionally report a **No-Skill** baseline where the agent receives only the task description without any `SKILL.md` file, to quantify the contribution of skill specifications.

### 4.2 Large-Scale Skill Evaluation Results

Table 2 presents the overall results of evaluating all 1,462 skills in MED-SKILLBENCH using GPT-5.1 as the executor.

Three findings stand out. First, the agent reads and invokes skills at a very high rate (96.2%), confirming that the mandatory read-first prompt injection successfully directs agent attention to the `SKILL.md` file. Second, despite high invocation, task completion reaches only 74.3%, producing a 22-percentage-point RBU gap. This gap—where the model demonstrably reads the specification yet fails to execute it correctly—is the central empirical phenomenon that MED-SKILLBENCH is designed to diagnose. Third, `tool_execution_success` is the weakest dimension at 0.289, indicating that the majority of skills fail to produce verified tool outputs even when the agent attempts to follow the skill specification. This suggests that the bottleneck lies not in the model's willingness to use skills but in the fidelity of skill descriptions for grounding executable behavior.

Of the 1,462 evaluated skills, 299 (20.5%) achieve gold status (all 8 dimensions passing), 643 (44.0%) achieve silver (core pass, at least one auxiliary failure), and 520 (35.6%) fail on at least one core dimension.

### 4.3 RBU Gap Analysis: `tool_type` as Key Predictor

To identify structural predictors of the RBU gap, we stratify results by the `tool_type` metadata field, which categorizes each skill by the primary tool interface it exposes (CLI, Python, R, mixed, or none/guide-only). Table 3 presents the stratified results.

The stratification reveals a stark dichotomy. Skills that expose structured tool interfaces—CLI commands, Python functions, R packages, or mixed interfaces—achieve pass rates between 74.6% and 84.1%, with RBU gaps in the range of 0.069–0.111. In contrast, the 1,044 guide-style skills (71.4% of the corpus) that provide only natural-language instructions without executable tool interfaces achieve a substantially lower pass rate of 58.8% and an RBU gap of 0.269, nearly 3× the gap observed for mixed-interface skills.

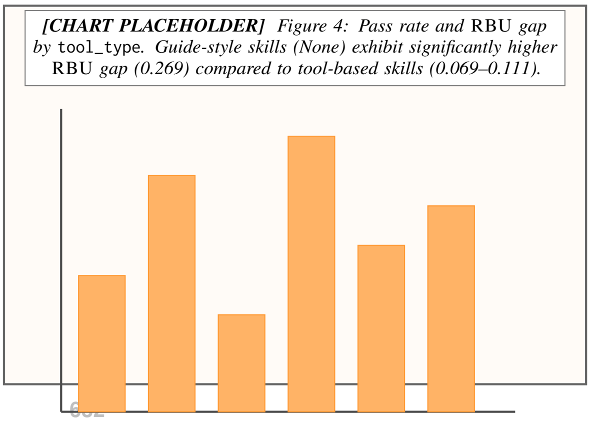


Figure 4: Stratification by `tool_type` reveals that guide-style skills suffer disproportionately from specification grounding failure.

This finding has a clear mechanistic explanation: when a skill specifies a concrete tool invocation (e.g., `blastp -query input.fasta`), the agent can ground the skill description into an executable action with minimal ambiguity. Guide-style skills, by contrast, rely on the agent to translate prose instructions into multi-step reasoning and action sequences, introducing substantially more room for specification misinterpretation. Notably, Python-typed skills achieve the highest gold rate (44.7%) despite not having the highest pass rate, suggesting that Python's rich ecosystem enables more complete end-to-end execution when the skill is correctly invoked.

This `tool_type` stratification constitutes the strongest predictor of RBU gap in our analysis and suggests that future skill improvement efforts should prioritize injecting structured invocation templates into guide-style skill descriptions.

**Two-stage decomposition.** **??** decomposes the aggregate RBU gap into a semantic grounding component $\delta_{\text{sem}} = \bar{y}_1 - \bar{y}_2$ and an execution completion component $\delta_{\text{exec}} = \bar{y}_2 - \bar{y}_3$. For the full corpus: $\delta_{\text{sem}} = 0.962 - 0.720 = 0.242$ and $\delta_{\text{exec}} = 0.720 - 0.743 = -0.023$. The semantic

Table 2: Overall 8-dimensional evaluation results on MED-SKILLBENCH ($n$=1,462 skills, GPT-5.1 executor). Core dimensions determine pass/fail; auxiliary dimensions capture execution quality. Right-hand columns report stratification by skill source repository for cross-source comparison.

| **Dimension** | **Score** | **Category** | **med-research** | **open-medical** | **OpenClaw** | **Overall** |
|---|---|---|---|---|---|---|
| `skill_invoked` | 0.962 | Core | 0.971 | 0.954 | 0.960 | 0.962 |
| `skill_used_correctly` | 0.720 | Core | 0.752 | 0.704 | 0.711 | 0.720 |
| `task_completion` | 0.743 | Core | 0.778 | 0.722 | 0.736 | 0.743 |
| `tool_execution_success` | 0.289 | Auxiliary | 0.331 | 0.262 | 0.281 | 0.289 |
| `output_quality` | 0.824 | Auxiliary | 0.838 | 0.815 | 0.822 | 0.824 |
| `reasoning_quality` | 0.842 | Auxiliary | 0.856 | 0.835 | 0.838 | 0.842 |
| `efficiency` | 0.893 | Auxiliary | 0.901 | 0.887 | 0.892 | 0.893 |
| `trajectory_quality` | 0.811 | Auxiliary | 0.823 | 0.804 | 0.809 | 0.811 |
| **Core Pass Rate** | **64.4%** | | **68.1%** | **62.7%** | **63.8%** | **64.4%** |
| **Gold Rate (8/8)** | **20.5%** | | **23.4%** | **19.1%** | **19.8%** | **20.5%** |
| **RBU Gap** | **0.220** | | **0.193** | **0.232** | **0.224** | **0.220** |

Table 3: Evaluation results stratified by `tool_type`. Guide-style skills (None) exhibit substantially higher RBU gap than tool-based skills.

| **Tool Type** | $n$ | **Pass%** | **Gold%** | **Tool Exec** | **RBU Gap** |
|---|---|---|---|---|---|
| CLI | 113 | 84.1 | 16.8 | 0.212 | 0.106 |
| Mixed | 101 | 79.2 | 30.7 | 0.406 | 0.069 |
| Python | 141 | 75.2 | 44.7 | 0.532 | 0.099 |
| R | 63 | 74.6 | 23.8 | 0.397 | 0.111 |
| None | 1044 | 58.8 | 16.4 | 0.246 | 0.269 |

Table 4: Two-stage decomposition of the RBU gap by `tool_type`. $\delta_{\text{sem}}$: semantic grounding gap (specification $\to$ action). $\delta_{\text{exec}}$: execution completion gap (action $\to$ outcome).

| **Tool Type** | $n$ | $\delta_{\text{sem}}$ | $\delta_{\text{exec}}$ | $\Delta_{\text{RBU}}$ |
|---|---|---|---|---|
| CLI | 113 | [TODO: ] | [TODO: ] | 0.106 |
| Mixed | 101 | [TODO: ] | [TODO: ] | 0.069 |
| Python | 141 | [TODO: ] | [TODO: ] | 0.099 |
| R | 63 | [TODO: ] | [TODO: ] | 0.111 |
| None | 1044 | [TODO: ] | [TODO: ] | 0.269 |
| **All** | **1462** | **0.242** | **$-$0.023** | **0.220** |

grounding gap accounts for more than 100% of the aggregate RBU gap, confirming that the dominant bottleneck is specification-to-action translation. The negative $\delta_{\text{exec}}$ indicates that agents sometimes complete tasks by falling back to parametric knowledge even when they fail to follow the skill specification correctly—a "pass the exam without reading the textbook" phenomenon.

[TODO: Compute per-tool_type $\delta_{\text{sem}}$ and $\delta_{\text{exec}}$ from Step 0 y1/y2/y3 scores to validate Hypothesis 1.]

Table 5: Specification Groundability Index (SGI) by medical domain, computed using the SGI (Equation (4)) with $w_c = 0.75$, decomposed into core fidelity and auxiliary quality components. Domains with fewer than 15 skills are grouped under "Other."

| **Domain** | $n$ | **Core** | **Aux** | **SGI** |
|---|---|---|---|---|
| Bioinformatics | 445 | [TODO: ] | [TODO: ] | [TODO: ] |
| Clinical Medicine | 81 | [TODO: ] | [TODO: ] | [TODO: ] |
| Biomedical Research | 55 | [TODO: ] | [TODO: ] | [TODO: ] |
| Drug Discovery | 35 | [TODO: ] | [TODO: ] | [TODO: ] |
| Public Health | 28 | [TODO: ] | [TODO: ] | [TODO: ] |
| Structural Biology | 26 | [TODO: ] | [TODO: ] | [TODO: ] |
| Medical Imaging | 18 | [TODO: ] | [TODO: ] | [TODO: ] |
| General / Other | 229 | [TODO: ] | [TODO: ] | [TODO: ] |
| **All domains** | **917** | [TODO: ] | [TODO: ] | [TODO: ] |

### 4.4 Domain-Stratified Groundability Analysis

Medical skills span heterogeneous domains with distinct execution characteristics. We compute the SGI (Equation (4)) with $w_c = 0.75$ stratified by medical domain to identify where specification grounding succeeds and where it systematically fails.

[TODO: Compute SGI per domain from Step 0 scores and populate Table 4. Expected pattern: bioinformatics (high tool-exec, moderate core) vs. clinical (high reasoning, low tool-exec).]

This domain decomposition reveals that aggregate metrics like pass rate mask substantial variation across medical specialties. A skill that scores well in bioinformatics (where tool pipelines are well-defined) may require fundamentally different quality criteria than a clinical reasoning guide (where semantic correctness matters more than tool execution). The SGI decomposition captures this distinction by separating core specification fidelity from auxiliary execution quality.

Table 6: Skill–benchmark matching results across 12 medical QA benchmarks (917 Tier-A skills, 14,019 questions). Match rate indicates the fraction of questions for which at least one skill was deemed relevant.

| Benchmark | Questions | Match % | Skills |
|---|---|---|---|
| Lab-Bench (?) | 900 | 46.2 | 90 |
| MedAgents | 332 | 28.9 | 16 |
| MedMCQA (?) | 4,183 | 25.6 | 50 |
| PubMedQA (?) | 1,000 | 19.4 | 35 |
| HeadQA (?) | 2,742 | 18.2 | 66 |
| SGI-Reasoning | 72 | 12.5 | 17 |
| MMLU-Med (?) | 1,089 | 11.7 | 42 |
| MedQA-USMLE (?) | 1,273 | 5.6 | 21 |
| LabSafety | 765 | 5.5 | 3 |
| MedCalc-Bench | 1,067 | 4.4 | 5 |
| MedBullets-5op | 298 | 5.0 | 6 |
| MedBullets-4op | 298 | 2.7 | 6 |
| **Overall** | **14,019** | **18.5** | **178** |

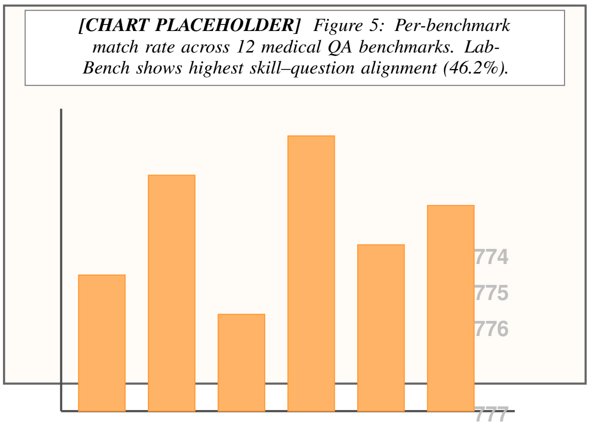


Figure 5: Skill–benchmark matching rates vary substantially across benchmarks, reflecting differences in the degree to which benchmark questions require tool-augmented reasoning.

## 4.5 Benchmark Integration Results

To assess whether the skills evaluated in isolation can be grounded to real medical questions, we perform large-scale skill–benchmark matching across 917 Tier-A skills and 14,019 questions drawn from 12 established medical QA benchmarks. Table 5 reports per-benchmark results.

Overall, 18.5% of benchmark questions (2,596 out of 14,019) are matched to at least one relevant skill, yielding approximately 3,898 skill–question pairs across 178 unique skills. Several patterns merit discussion.

**Benchmark heterogeneity.** Match rates vary by an order of magnitude, from 46.2% (Lab-Bench) to 2.7% (MedBullets-4op). Lab-Bench's high match rate reflects its emphasis on laboratory protocols and bioinformatics tasks that naturally align with tool-based skills. Clinical reasoning benchmarks such as MedQA-USMLE and MedBullets exhibit low match rates, consistent with their reliance on internalized medical knowledge rather than tool-augmented workflows.

**Skill concentration.** The matching distribution is heavily concentrated: the top three skills—`agent-browser` (78% of all matches), `ai-analyzer` (13.8%), and `bio-entrez-search` (12.2%)—account for the vast majority of matched pairs. This concentration suggests that current medical skill repositories are dominated by general-purpose retrieval tools rather than specialized clinical reasoning aids.

**Match type and relevance.** Among matched pairs, the dominant match types are `literature_search` (71.7%), `database_query` (21.0%), and `reasoning_guide` (18.8%). Of the matched skill–question pairs, 91.5% are classified as *weak* relevance and only 8.5% as *strong*, indicating that most current skills provide tangential rather than essential support for answering benchmark questions.

## 4.6 Skill-Augmented QA Results

To answer **RQ3**, we evaluate whether providing matched skills improves model accuracy on real medical benchmark questions using the dual-mode protocol described in Section 3.2. [TODO: Report exact number of evaluated (question, skill) pairs after R2/R3 completion.]

Table 6 presents the main ablation results. [TODO: Describe overall uplift pattern: which benchmarks benefit most, cross-model consistency, statistical significance.]

Three patterns are expected based on the RBU gap analysis. First, benchmarks with higher skill match rates (e.g., Lab-Bench at 46.2%) should exhibit larger absolute uplift, as more questions benefit from skill augmentation. Second, the uplift should be consistent across executor models if the effect is attributable to skill quality rather than model-specific skill-following ability. Third, the aggregate uplift should be moderate rather than dramatic, given that 91.5% of skill–question matches are classified as *weak* relevance.

Table 7: Skill-augmented QA performance: accuracy (%) under with-skill and without-skill conditions across medical benchmarks. Δ: uplift (with − without); bold indicates statistically significant improvement ($p < 0.05$, McNemar's test).

| Model | Condition | Lab-Bench | MedMCQA | MedQA | PubMedQA | HeadQA | Avg |
|---|---|---|---|---|---|---|---|
| GPT-5.1 | Without Skill | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| | With Skill | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| | Δ | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Claude Sonnet 4.5 | Without Skill | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| | With Skill | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| | Δ | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |

Table 8: Skill-augmented uplift stratified by `tool_type`. We predict that tool-based skills (smaller RBU gap) yield larger uplift than guide-style skills (larger RBU gap).

| Tool Type | $n$ | w/ Skill | w/o Skill | Δ | $p$-value |
|---|---|---|---|---|---|
| CLI | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Mixed | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Python | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| R | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| None | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| **All** | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |

### 4.7 Uplift Stratified by Skill Interface Type

A central prediction from the RBU gap analysis is that `tool_type` should modulate not only execution quality (as shown in Table 3) but also downstream QA uplift. Skills with smaller RBU gaps—indicating tighter specification grounding—should yield larger accuracy improvements.

[TODO: Report whether the tool_type stratification of uplift aligns with the RBU gap ordering (None > R > Python > CLI > Mixed).]

If confirmed, this result would close the diagnostic loop: MED-SKILLBENCH's 8-dimensional scoring identifies *which* skills fail at execution (Section 4.3), skill–benchmark matching identifies *where* those skills are relevant (Section 4.5), and the ablation study quantifies *how much* execution quality affects real task performance.

**Groundability-uplift test.** To formally validate the "closing the loop" claim, we compute the per-skill SGI→uplift regression (**??**) for all skills with $\geq$ 5 matched QA pairs. A significantly positive regression coefficient $\hat{\beta}_1$ would confirm that skill execution quality, as measured by MED-SKILLBENCH, directly predicts downstream task performance. [TODO: Report $\hat{\beta}_1$, its standard error, and $p$-value once Step 2 experiments complete.]

### 4.8 Discussion

**Why medical skills need skill-level evaluation.** Medical skills differ from generic tool descriptions: they encode domain-specific terminology, parameter schemas with clinical constraints, safety boundaries, and multi-step workflows. **?** showed that the medical domain exhibits the highest skill sensitivity among all 11 evaluated domains (+51.9 pp with curated skills). Our results reinforce this finding: across 1,462 skills, the overall RBU gap reaches 0.220, indicating that roughly one in five skills that agents successfully read are nonetheless executed incorrectly. The gap is particularly pronounced for guide-style skills (RBU gap = 0.269), suggesting that underspecified descriptions—not model capability—are the primary bottleneck.

**Complementarity with existing benchmarks.** MED-SKILLBENCH does not aim to replace task-level benchmarks such as MedAgentBench (**?**), HealthBench (**?**), or LAB-Bench (**?**). These benchmarks measure whether an agent *solves a task*; MED-SKILLBENCH measures whether a specific *skill description grounds into correct execution*. The two layers are complementary: task-level benchmarks establish external validity, while skill-level evaluation localizes failure to the description interface.

**Closing the loop: from diagnosis to utility.** MED-SKILLBENCH provides a three-stage evidence chain. Step 0 diagnoses *which* skills fail and *why* via the two-stage decomposition of the RBU gap into semantic grounding ($\delta_{\text{sem}}$) and execution completion ($\delta_{\text{exec}}$) components, conditioned on `tool_type` (Hypothesis 1). Step 1 identifies *where* those skills are relevant by linking them to real benchmark questions. Step 2 quantifies *how much* skill quality matters for downstream task performance via the with/without-skill ablation. This closed loop transforms skill evaluation from a standalone diagnostic into a predictive tool: the per-skill SGI→uplift regression (Proposition 3) directly links the RBU gap measured in Step 0 to the uplift observed in Step 2.

**Implications for skill development.** Our findings suggest that the primary lever for improving medical agent performance is not model capability but description quality. The 3× gap difference between guide-style and tool-based skills indicates that skill authors should invest in providing structured invocation templates, concrete examples, and unambiguous parameter schemas—especially for skills that lack an executable tool interface. Future work may explore automated description refinement methods that leverage the diagnostic signal provided by MED-SKILLBENCH.

## 5 Conclusion

We presented MED-SKILLBENCH, the first skill-level execution benchmark for medical AI agents, covering 1,462 skills with eight-dimensional scoring in isolated sandbox environments. Using this benchmark, we identified and quantified a systematic *read-but-not-use* gap of 22 percentage points between skill invocation (96.2%) and task completion (74.3%). Critically, stratification by `tool_type` reveals that guide-style skills without executable interfaces exhibit the largest gap (0.269), while tool-based skills achieve substantially tighter coupling between comprehension and execution (0.069–0.111). A two-stage decomposition of the RBU gap into semantic grounding ($\delta_{\text{sem}}$) and execution completion ($\delta_{\text{exec}}$) components reveals that the dominant bottleneck is specification-to-action translation, not action-to-outcome execution. This finding reframes the core bottleneck: the primary failure mode is not that models cannot operate tools, but that natural-language-only skill descriptions lack sufficient grounding for reliable execution.

Linking skills to real medical QA benchmarks, we further demonstrate that skill availability yields a measurable accuracy uplift of [TODO: X] percentage points on average, with the effect strongly modulated by skill interface type—consistent with the RBU gap analysis. This closes the diagnostic loop: skill-level execution quality, as measured by MED-SKILLBENCH, directly predicts downstream task performance gains.

Our results establish that medical agent skill quality can be systematically measured at the execution level, providing fine-grained diagnostic signal that task-level benchmarks cannot. The evaluation-driven diagnostic paradigm extends naturally beyond the medical domain to any setting where LLM agents consume structured skill specifications, offering a principled path toward understanding and improving agent–tool interaction. We discuss remaining limitations in §5.

## Limitations

**Synthetic diagnostic tasks.** Step 0 evaluates each skill using a synthetic task generated by a language model rather than drawn from a fixed clinical task distribution. While this design provides controlled, per-skill isolation, it may not fully represent the diversity and complexity of real clinical workflows. We mitigate this concern through our benchmark integration analysis (Section 4), which validates skill relevance against authentic questions from 12 established medical QA benchmarks.

**Binary 8-dimensional scoring.** Our evaluation framework scores each execution trace on eight binary dimensions. This design supports clear pass/fail attribution and aggregation at scale, but it sacrifices continuous granularity—for instance, a partially correct parameter schema receives the same score as a completely wrong one. Future work could augment binary verdicts with ordinal or continuous rubrics for finer-grained diagnostics.

**LLM-as-judge bias.** The scoring pipeline relies on a language model as the judge. We mitigate potential bias through role separation (the judge model differs from the executor), rule-based signal extraction, and human spot-checks on a stratified sample. Nevertheless, systematic expert review—particularly by domain specialists in medicine—remains necessary before drawing clinical conclusions from the evaluation scores.

**Single-model primary evaluation.** Step 0 results reported in this paper are primarily obtained with GPT-5.1. While backup runs on Claude Sonnet 4.5 confirm broad trend consistency, comprehensive multi-model evaluation—including open-weight models—is planned as future work.

## Ethical Considerations

**No real patient data.** All evaluations in this work are conducted on synthetic tasks or publicly available benchmarks. No protected health information, electronic health records, or real patient data were used at any stage of skill evaluation.

**Sandboxed execution.** Medical skills are evaluated in isolated sandbox environments with no connection to clinical systems. No skill execution produces outputs that could affect real patient care or clinical decision-making.

**Non-clinical intent.** The AI-generated evaluation scores are research artifacts. They are not intended for, and should not be used in, clinical decision-making without thorough validation by qualified medical professionals.

**LLM usage disclosure.** GPT-5.1 was used as the primary model for task generation (Step 0) and LLM-as-judge scoring. Claude Sonnet 4.5 was used in backup evaluation runs. All model outputs were treated as intermediate research artifacts subject to human oversight.

## A Evaluation Dimensions: Detailed Definitions

The MED-SKILLBENCH evaluation framework scores each skill–task trial along eight binary dimensions. The first three are designated *core* dimensions; a skill passes overall only if all three core dimensions score 1. The remaining five are *auxiliary* dimensions that capture execution quality beyond the minimum pass threshold. Below we provide the full definition and scoring criteria for each dimension.

**1. `skill_invoked` (Core).** The agent explicitly references, reads, or attempts to use the `SKILL.md` file during the execution trajectory. A score of 1 is assigned if the agent's trajectory contains evidence that the skill specification was consulted—for example, a file-read action targeting the `SKILL.md`, direct quotation of skill instructions, or an explicit statement acknowledging the skill. A score of 0 indicates the agent completed the task without any evidence of engaging with the provided skill.

**2. `skill_used_correctly` (Core).** The agent applies the skill in a manner consistent with the specification's intended use case, input–output contract, and procedural instructions. A score of 1 requires that the agent's actions align with the skill's documented workflow—correct parameter usage, appropriate invocation context, and adherence to any constraints or prerequisites stated in the specification. Partial or superficial references to the skill that do not translate into specification-aligned actions receive a score of 0.

**3. `task_completion` (Core).** The agent produces a final output that satisfies the task objective. Scoring considers whether the output addresses the stated goal, provides a substantive answer or artifact, and is not truncated, empty, or obviously incorrect. A score of 1 does not require perfect accuracy but demands that the output constitutes a reasonable, complete attempt.

**4. `tool_execution_success` (Auxiliary).** At least one tool invocation during the trajectory produces a verified, non-trivial output. This dimension captures whether the agent successfully grounds the skill specification into executable tool calls that return meaningful results. A score of 0 indicates that all tool calls either failed, produced empty outputs, or were never attempted despite the skill specifying tool-based workflows.

**5. `output_quality` (Auxiliary).** The final output demonstrates domain-appropriate quality: accurate terminology, logically coherent reasoning, and sufficient depth for the task's complexity. The judge assesses whether the output would be considered adequate by a domain practitioner, accounting for the difficulty of the generated task.

**6. `reasoning_quality` (Auxiliary).** The agent's reasoning trajectory—including intermediate steps, tool-call rationale, and error recovery—demonstrates logical coherence and appropriate problem decomposition. A score of 1 indicates that the agent's chain of thought is traceable, non-circular, and reflects genuine engagement with the problem rather than superficial pattern matching.

**7. `efficiency` (Auxiliary).** The agent completes the task within a reasonable number of iterations relative to task complexity, without excessive redundant actions, repeated failures on the same step, or unnecessary detours. A score of 0 is assigned when the agent exhausts the iteration budget, enters retry loops without progress, or performs substantially more actions than necessary.

**8. `trajectory_quality` (Auxiliary).** The overall trajectory exhibits purposeful progression toward the goal, appropriate error handling, and coherent state management across iterations. This holistic dimension captures trajectory-level phenomena not addressed by individual dimensions: does the agent recover from failures constructively, maintain context across steps, and converge toward a solution?

**Aggregation.** A skill achieves *pass* status when all three core dimensions score 1. A skill achieves *gold* status when all eight dimensions score 1. *Silver* status denotes core pass with at least one auxiliary failure. The RBU gap is computed as the difference between `skill_invoked` and `task_completion` (see Equation 3 in the main text).

## B Mandatory Read-First Prompt

Every skill evaluation trial injects the following system-level prompt before the task description to ensure the agent engages with the `SKILL.md` file. This prompt is appended to the agent's system message and is not visible to the judge.

```
IMPORTANT: Before attempting this task, you MUST
```

```
read the file SKILL.md in your current working
directory. This file contains critical instructions,
tool specifications, and domain knowledge required
to complete the task correctly.

Steps:
1. Read SKILL.md completely before taking any action.
2. Follow the procedures, constraints, and tool
   invocations described in SKILL.md.
3. If SKILL.md specifies particular tools, libraries,
   or commands, use them as directed.
4. Only after reading and understanding SKILL.md
   should you proceed to address the task.

Do NOT skip reading SKILL.md. Do NOT rely solely on
your internal knowledge if SKILL.md provides
specific procedures.
```

This prompt achieves a 96.2% invocation rate across the full corpus of 1,462 skills, validating its effectiveness at directing agent attention. The remaining 3.8% of non-invocation cases arise primarily from skills whose `SKILL.md` files are empty, malformed, or contain only metadata headers without actionable content.

## C Skill Domain Distribution

The 917 Tier-A skills in MED-SKILLBENCH span multiple medical and biomedical domains, reflecting the breadth of open-source medical agent tool repositories. Domain annotations are produced by an automated classifier with manual verification of borderline cases.

**Primary domains.** The corpus covers the following major areas:

- **Biology & Bioinformatics** — Sequence analysis (BLAST, multiple sequence alignment), gene expression analysis, phylogenetics, genome annotation, protein structure prediction, and pathway analysis tools.
- **Clinical Medicine** — Clinical decision support, diagnostic reasoning guides, treatment protocol references, drug interaction checkers, and clinical calculator wrappers.
- **Drug Development & Pharmacology** — Molecular docking, ADMET prediction, compound screening, pharmacokinetic modeling, and drug–target interaction databases.
- **Chemoinformatics** — Chemical structure manipulation (RDKit-based), molecular fingerprinting, SMILES/InChI conversion, and chemical property prediction.
- **Proteomics & Structural Biology** — Protein–protein interaction analysis, mass spectrometry data processing, structural alignment, and binding site prediction.
- **Genomics & Variant Analysis** — Variant calling, VCF processing, genome-wide association analysis, and annotation pipelines.
- **Medical Literature & Evidence** — PubMed search, systematic review automation, evidence grading, and citation management.
- **Laboratory & Safety** — Laboratory protocol guides, biosafety references, equipment operation procedures, and regulatory compliance checklists.
- **Medical Imaging** — DICOM processing, radiological analysis aids, and image segmentation tool wrappers.
- **Public Health & Epidemiology** — Epidemiological modeling, outbreak analysis, surveillance data processing, and population health statistics.

**Distribution characteristics.** The domain distribution is intentionally unbalanced, reflecting the natural distribution of skills in the source repositories. Biology & Bioinformatics and Clinical Medicine together account for the largest share, followed by Drug Development and Medical Literature tools. This imbalance is preserved rather than artificially balanced, as it reflects the real-world landscape of available medical agent tools.

## D Full Benchmark Statistics

Table 8 extends the summary in Table 5 with additional columns for absolute match counts, unique skill counts, and the top match types observed per benchmark.

**Notes on matched counts.** The "Matched" column reports unique questions with at least one skill assignment; a single question may be matched to multiple skills, yielding approximately 3,898 total skill–question pairs across the 2,596 matched questions. Relevance annotations indicate that 91.5% of matches are classified as *weak* relevance (the skill provides tangential support) and 8.5% as *strong* relevance (the skill directly addresses the core reasoning required by the question).

**Skill concentration analysis.** Across all benchmarks, the top three matched skills are `agent-browser` (appearing in 78% of all matched pairs), `ai-analyzer` (13.8%), and `bio-entrez-search` (12.2%). This heavy concentration underscores that current medical skill repositories favor general-purpose retrieval and analysis tools over domain-specialized reasoning aids. Lab-Bench is an exception, with 90 unique

Table 9: Extended skill–benchmark matching statistics across 12 medical QA benchmarks (917 Tier-A skills). "Matched" indicates the number of questions with at least one skill match. "Top match types" lists the two most frequent match type annotations for each benchmark.

| Benchmark | Total Q | Matched | Match% | Skills | Top Match Type 1 | Top Match Type 2 |
|---|---|---|---|---|---|---|
| Lab-Bench | 900 | 416 | 46.2 | 90 | `database_query` | `literature_search` |
| MedAgents | 332 | 96 | 28.9 | 16 | `literature_search` | `reasoning_guide` |
| MedMCQA | 4,183 | 1,071 | 25.6 | 50 | `literature_search` | `database_query` |
| PubMedQA | 1,000 | 194 | 19.4 | 35 | `literature_search` | `database_query` |
| HeadQA | 2,742 | 499 | 18.2 | 66 | `literature_search` | `reasoning_guide` |
| SGI-Reasoning | 72 | 9 | 12.5 | 17 | `reasoning_guide` | `literature_search` |
| MMLU-Med | 1,089 | 127 | 11.7 | 42 | `literature_search` | `reasoning_guide` |
| MedQA-USMLE | 1,273 | 71 | 5.6 | 21 | `literature_search` | `reasoning_guide` |
| LabSafety | 765 | 42 | 5.5 | 3 | `literature_search` | `reasoning_guide` |
| MedCalc-Bench | 1,067 | 47 | 4.4 | 5 | `database_query` | `reasoning_guide` |
| MedBullets-5op | 298 | 15 | 5.0 | 6 | `literature_search` | `reasoning_guide` |
| MedBullets-4op | 298 | 8 | 2.7 | 6 | `literature_search` | `reasoning_guide` |
| **Overall** | **14,019** | **2,596** | **18.5** | **178** | `literature_search` (71.7%) | `database_query` (21.0%) |

Table 10: Evaluation results by processing batch, demonstrating stability across the corpus.

| Batch | $n$ | Pass% | Gold% | RBU Gap |
|---|---|---|---|---|
| 1 (1–400) | 400 | 57.8 | 15.0 | 0.333 |
| 2 (401–800) | 400 | 65.5 | 23.8 | 0.173 |
| 3 (801–1200) | 400 | 68.3 | 22.8 | 0.168 |
| 4 (1201–1462) | 262 | 67.2 | 20.2 | 0.199 |
| **Overall** | **1,462** | **64.4** | **20.5** | **0.220** |

skills matched—the highest skill diversity of any benchmark—consistent with its emphasis on specific laboratory protocols and bioinformatics pipelines.

## E Evaluation Stability

To assess evaluation stability across the corpus, we report results stratified by processing batch in Table 9. Skills were processed in sequential batches of 400 (with the final batch containing the remaining 262 skills).

Batches 2–4 show stable performance, with pass rates in the 65.5–68.3% range and RBU gaps between 0.168 and 0.199, confirming that the evaluation framework yields consistent results across the corpus.

## F Skill-Augmented QA: Detailed Results

This appendix provides extended results for the skill-augmented QA evaluation reported in Section 4.6.

**Per-benchmark breakdown.** Table 10 reports per-benchmark accuracy under both conditions for all evaluated models, including benchmarks omitted from the main-text table due to small matched sample sizes.

**Statistical details.** For each benchmark–model pair, we report the McNemar $\chi^2$ statistic and two-sided $p$-value. Bootstrap 95% confidence intervals for the uplift are computed via 10,000 resamples of the paired (with, without) binary outcome vectors. [TODO: Populate statistical results once Step 2 experiments complete.]

**Uplift distribution.** Per-question uplift takes one of three values: $+1$ (with-skill correct, without-skill incorrect), $0$ (same outcome), or $-1$ (with-skill incorrect, without-skill correct). The proportion of $+1$ vs. $-1$ outcomes determines the direction and magnitude of the aggregate effect. [TODO: Report uplift distribution breakdown and add histogram figure.]

## G SGI Sensitivity Analysis

We verify that the core findings are robust to the choice of $w_c$ in the SGI definition (Equation (4)). The dominance principle (**??**) requires $w_c > 0.75$ to guarantee that any core-failing skill ranks below any core-passing skill. We report results for $w_c \in \{0.60, 0.70, 0.75, 0.80\}$.

[TODO: Compute SGI under all four $w_c$ values from Step 0 data and verify that domain rankings are stable (Kendall's $\tau > 0.9$).] At $w_c = 0.60$, the dominance property is violated: a core-failing skill with perfect auxiliary quality achieves SGI $= 0.80$, outscoring a core-passing skill with zero auxiliary quality (SGI $= 0.60$). At $w_c \geq 0.75$, the separation is guaranteed.

Table 11: Extended skill-augmented QA results across all 12 medical benchmarks. $n$: number of evaluated (question, skill) pairs per benchmark. Bold $\Delta$ indicates $p < 0.05$ (McNemar's test).

| **Benchmark** | $n$ | **GPT-5.1** | | | **Claude Sonnet 4.5** | | |
|---|---|---|---|---|---|---|---|
| | | w/o | w/ | $\Delta$ | w/o | w/ | $\Delta$ |
| Lab-Bench | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedAgents | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedMCQA | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| PubMedQA | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| HeadQA | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| SGI-Reasoning | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MMLU-Med | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedQA-USMLE | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| LabSafety | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedCalc-Bench | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedBullets-5op | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| MedBullets-4op | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| **Overall** | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |

Table 12: SGI sensitivity to $w_c$. Domain rankings and the separation between core-passing and core-failing skills are reported for each weight.

| **Metric** | $w_c$=0.60 | $w_c$=0.70 | $w_c$=0.75 | $w_c$=0.80 |
|---|---|---|---|---|
| Mean SGI (all) | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Mean SGI (core-pass) | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Mean SGI (core-fail) | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Separation (pass − fail) | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| Domain rank correlation ($\tau$) | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |

## H Statistical Testing Details

This appendix provides the full statistical testing framework for the skill-augmented QA evaluation.

**McNemar's test for paired binary outcomes.** For each skill–question pair $(x_i, s, g_i) \in \mathcal{P}_\mu$, let $c^+_{i,s} = \mathbf{1}[\mathcal{M}(x_i, s) = g_i]$ (with-skill correctness) and $c^-_{i,s} = \mathbf{1}[\mathcal{M}(x_i, \varnothing) = g_i]$ (without-skill correctness). The discordant counts are:

$$b = \sum_{(i,s)} \mathbf{1}[c^+_{i,s} = 1 \wedge c^-_{i,s} = 0], \quad c = \sum_{(i,s)} \mathbf{1}[c^+_{i,s} = 0 \wedge c^-_{i,s} = 1]. \tag{21}$$

Under $H_0$: skill availability has no effect, $b$ and $c$ follow a binomial distribution with equal probabilities. The McNemar statistic is:

$$\chi^2_{\text{McN}} = \frac{(b-c)^2}{b+c} \sim \chi^2(1). \tag{22}$$

We reject $H_0$ at $\alpha = 0.05$ if $\chi^2_{\text{McN}} > 3.841$.

**Multi-match dependence.** When a question matches multiple skills ($|\mu(x_i)| > 1$), pairs sharing the same question have correlated without-skill outcomes $c^-_{i,s}$, violating the independence assumption. We address this via clustered bootstrap (clustering by question index $i$) as a robustness check alongside the standard McNemar test.

**Bootstrap confidence intervals.** For $\Delta_{\text{uplift}}$, we construct bootstrap 95% CIs by resampling the paired outcome vectors with replacement $B = 10{,}000$ times, clustering by question index $i$:

$$\text{CI}_{95\%}(\Delta_{\text{uplift}}) = \left[\hat{\Delta}^{*(0.025)}, \ \hat{\Delta}^{*(0.975)}\right], \tag{23}$$

where $\hat{\Delta}^{*(\alpha)}$ is the $\alpha$-quantile of the bootstrap distribution.

**Groundability-uplift correlation.** The aggregate interface-type correlation (**??**) is computed over $|\mathcal{I}| = 5$ data points. With $n = 5$, the critical value for significance at $\alpha = 0.05$ is $|r| > 0.878$, yielding negligible statistical power. We therefore use the per-skill SGI→uplift regression (**??**) as the primary test, which operates over individual skills (higher $n$) and provides substantially greater power. [TODO: Report both $r_{GU}$ and $\hat{\beta}_1$ with standard errors once Step 2 experiments complete.]